

Stake.Link Metis Staking Audit Summary

Prepared by Cyfrin

Version 2.0

Lead Auditors

0kage

Hans

November 18, 2024

Contents

1 About Cyfrin 2

2 About This Document 2

3 Disclaimer 2

4 Copyright and License 2

5 Audit Overview 3

6 Audit Scope 4

7 Risk Classification 4

8 Protocol Summary 4

9 Attestation 5

1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

2 About This Document

This document is a public summary of the security audit Cyfrin performed for Stake.Link on Stake.Link Metis Staking. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number and final status of findings by severity. It closes with Cyfrin's attestation of the review's scope and final remediation status.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 2.0 dated November 18, 2024, was delivered to Stake.Link, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

4 Copyright and License

© 2026 Cyfrin Inc. All rights reserved. Material from Stake.Link or third parties remains the property of its owners.

This summary may be shared only in complete and unmodified form, so that any copy can be checked against the hash in its attestation. Short quotations with attribution to Cyfrin are permitted, provided they do not misrepresent this summary's scope or results. Except as permitted by applicable law, no part of this summary may otherwise be copied, modified, adapted, sold or sublicensed, or used to create derivative works, without Cyfrin's prior written permission.

Cyfrin and the Cyfrin logo are trademarks of Cyfrin Inc. and may not be used to suggest that Cyfrin has reviewed or endorses anything outside this summary's scope, or to misrepresent its results.

5 Audit Overview

Over the course of 10 days, the Cyfrin team conducted an audit on the Stake.Link Metis Staking code provided by Stake.Link. The findings consist of 3 High, 2 Medium & 1 Low severity issues.

Summary

Project Name	Stake.Link Metis Staking
Repository	contracts
Commit	06de237d7785...
Audit Timeline	Oct 14th - Oct 25th, 2024
Methods	Manual Review, Stateful Fuzzing

Issues Found

Severity	Found	Resolved	Acknowledged
Critical Risk	0	0	0
High Risk	3	2	1
Medium Risk	2	2	0
Low Risk	1	0	1
Informational	0	0	0
Gas Optimizations	0	0	0
Total	6	4	2

6 Audit Scope

The audit focused on changes introduced in PR #128 implementing the Metis liquid staking protocol's core contracts. The scope included cross-chain communication, strategy management, and sequencer vault implementations.

Changes in the following files were part of the current audit scope

```
L1Strategy.sol
L2Strategy.sol
SequencerVault.sol
L1Transmitter.sol
L2Transmitter.sol
PriorityPool.sol
WithdrawalPool.sol
IL1StandardBridge.sol
IL1StandardBridgeGasOracle.sol
IL1Strategy.sol
IL2StandardBridge.sol
IL2StandardBridgeGasOracle.sol
IL2Strategy.sol
IMetisLockingInfo.sol
IMetisLockingPool.sol
```

7 Risk Classification

	Impact: High	**Impact: Medium**	**Impact: Low**
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

8 Protocol Summary

The codebase implements a liquid staking solution for Metis sequencers, operating across Ethereum (L1) and Metis (L2) chains. Users can deposit tokens on L2 and receive liquid staking tokens (LST) in return, while their deposits are utilized for sequencer staking on L1.

Key Components

Following are the key contracts on the Metis (L2) chain:

- L2Strategy: Manages deposits and LST minting on Metis
- WithdrawalPool: Handles queued withdrawals in FIFO order
- PriorityPool: Controls deposit flow and LST distribution

Following are the key contracts on Ethereum (L1) chain:

- L1Strategy: Manages sequencer vault creation and staking on Ethereum
- SequencerVault: Individual vaults for Metis sequencer staking
- MetisLockingPool: Interface with Metis sequencer staking system

Following are the contracts responsible for cross chain communication:

- L1Transmitter/L2Transmitter: Handle cross-chain messaging via CCIP and native token bridging via L1/L2 Standard Bridges. Both these contracts manage state synchronization between L1 and L2

9 Attestation

Cyfrin attests that it performed a security review of the Stake.Link Metis Staking code provided by Stake.Link.

The repositories, commits, review period and findings with their final status are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at https://github.com/Cyfrin/cyfrin-audit-reports. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2024-11-18-cyfrin-stake.link-metis-staking-v2.0.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to Stake.Link has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=de6ff47e18aa2ceaf62cf297deb044aa2059adbdfe66fd8e9c0cda2f639669a3
F=2024-11-18-cyfrin-stake.link-metis-staking-v2.0-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.